UNESCO Institute for Statistics

Talking to Our Data

An MCP Server for Natural Language SDG4 Indicator Queries

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It always starts the same way

"Can you pull the literacy coverage and trend for these 15 countries for the next regional report?"

1. Locate indicator IDs
2. Filter bulk download
3. Check MAGNITUDE flags
4. Build chart in R / Stata
5. Write summary

× 15 countries × every reporting cycle

From chatbot guesses to verified tool calls

Without MCP

- AI answers from training data
- Data may be outdated or wrong
- Indicator codes invented (hallucinated)
- No connection to our live database
- No validity filtering applied

VS

With MCP Server

- AI calls validated tools
- Data fetched live from UIS API
- Indicator IDs from verified codebook
- Validity filter applied server-side
- AI writes narrative, server owns numbers

How It Works

What happens when you type a natural language query

1

resolve_country

"Qatar" → QAT
ISO3, region, population

2

resolve_sdg_ indicators

"4.6.1" → LR.AG15T99
LR.AG15T24 … (codebook)

3

get_indicator_ data

Live UIS API call
Validity filter applied

4

format_country_ briefing

Gap years, YoY change
post-2020 status

5

AI narrative

Neutral, factual text
Style guide enforced

User: "Generate a data briefing for Qatar covering basic literacy indicators."

Previous Version — What Went Wrong

Three problems in the earlier prototype — and how they were fixed

⚠ Hallucinated indicator IDs

AI guessed indicator codes that don't exist in our database.

✓ Fix

All IDs now resolved from a versioned codebook derived from the UIS SDG4 Monitoring Framework file. Pre-flight validation raises an error if any ID is missing.

⚠ Modelled estimates as reported data

LR.GALP.* projections were returned alongside observed rates with no distinction.

✓ Fix

Validity filter applied server-side: MAGNITUDE ∈ {SUPP, NA, INCLUDED} excluded; CR.MOD.* and LR.GALP.* prefixes blocked before the AI sees any data.

⚠ Country names not recognised

"Tanzania", "South Korea", "UAE" returned nothing. ISO3 had to be typed manually.

✓ Fix

Bundled alias dictionary: 273 name variants for all 214 UIS World countries. "Tanzania" → TZA, "South Korea" → KOR, "Iran" → IRN, etc.

What it can do for you

Country briefing

Coverage, trend, gap years, and latest value for any indicator and country — in one natural language query.

Coverage analysis

Post-2020 data availability across 214 UIS countries. Country count and % of world population covered.

Trend & gap detection

Year-over-year changes computed automatically. Missing years flagged. Latest data point identified.

Indicator resolution

Type an SDG number — 4.6.1, 4.7.1 — and the correct indicator IDs are retrieved from the codebook instantly.

Validity filtering

Observed data only. MAGNITUDE exclusions and modelled-estimate prefixes blocked server-side automatically.

Regional breakdown

List countries by UIS sub-region. Compare coverage across Sub-Saharan Africa, Arab States, Central Asia, etc.

LIVE DEMO

1. *List all global SDG 4 indicators.*
2. *Resolve country name "South Korea".*
3. *Generate a data briefing for Qatar covering basic literacy indicators.*
4. *Check data availability for Uzbekistan on SDG 4.1.2 after 2020.*

Queries 1–2 use codebook only (offline). Queries 3–4 call the UIS API.

Value Added

Traditional workflow vs. MCP server

Step	Traditional	With MCP Server
Specify indicator	Must know exact indicator ID	Type SDG number (e.g. 4.6.1)
Locate country	Must know ISO3 code	Type country name in any language variant
Get data	Filter bulk download CSV manually	Live API call, automatic
Validity check	Check MAGNITUDE column manually	Applied server-side, invisible to user
Compute coverage	Script in R or Stata	Returned in the same query
Build chart	R / Stata / Excel	Generated on request, inline
Write summary	Manual — 15–30 min per country	Generated in seconds, neutral style enforced
Time per country profile	**30–60 minutes**	**2–3 minutes**

What it does not replace

Gaps reflect the data, not the tool

If a country has not reported, the tool shows no data. Non-technical users may not distinguish data absence from indicator non-applicability. Context is always needed.

Codebook needs manual updates

When UIS publishes a new indicator framework version, the codebook JSON must be updated and the server redeployed. This is a known manual step — it is not automated.

API network dependency

Data fetch tools require access to api.uis.unesco.org. On some UNESCO network configurations or from certain regions, access may be restricted. Codebook tools work offline.

Writing style requires review

The server includes a style guide enforcing neutral language. AI outputs intended for external use should still be reviewed — subtle evaluative phrasing can slip through.

Not a substitute for analytical judgement

The tool accelerates data retrieval and formatting. It does not replace expertise in data quality assessment, indicator selection, or cross-country comparability.

Possible extensions

Automated release reports

Coverage summaries triggered automatically after each UIS data release cycle.

Formatted Word / PDF export

Country profile documents generated directly in UIS report template format.

Multi-indicator comparison

Compare coverage across a custom indicator basket for a user-defined country group.

Indicator spec auto-update

Script to regenerate the codebook from the Framework Excel file after a new release.

Internal API extension

Connect to UIS internal systems and non-public datasets beyond the public API.

Other UIS datasets

Extend beyond SDG4 — EMIS, PISA, TALIS, or UIS finance indicators.

What you need to use it

Prerequisites

- Python 3.10+ (Anaconda works)
- VS Code
- GitHub Copilot or Claude extension
- Access to the GitHub repository
- Internet access to api.uis.unesco.org

To get started

1. Clone the GitHub repository
2. conda activate base
3. pip install "mcp[cli]" requests pandas
4. python server.py --test (10 checks)
5. Configure .vscode/mcp.json
6. Open Copilot Chat in Agent mode

No coding required to use it once configured. The server runs in the background.

Questions & Discussion

Server owns the numbers

Every value comes from the UIS API — never from the AI's memory.

Codebook is the safeguard

Indicator IDs are always validated, never guessed.

You don't need to code

Type in natural language. The tools handle everything else.

https://github.com/yifnli/uis-sdg4-mcp https://github.com/yifnli/uis-sdg4-mcp